Unitarization Methods: Newton, Halley, and Alternatives

Overview

Unitarization refers to projecting a general matrix onto the unitary group $U(N)$, or special unitary group $SU(N)$. Several iterative methods exist: Newton's method, Halley's method, Cayley-Hamilton projection, and exponential projection.

1 Newton's Method

Goal: Given a non-unitary matrix X, find a nearby unitary matrix $U \approx X$ minimizing $\|U - X\|$.

Based on: Polar decomposition: $X = UH$, where U is unitary and H is Hermitian positive-definite.

Iterative Update:

$$F = X^\dagger X, \quad W = \frac{\mathrm{Tr}(F)}{\|F\|^2}, \quad X \leftarrow \sqrt{W} X$$
$$R = I - F$$

- If $\|R\| \ll 1$:
$$X \leftarrow X + 0.5XR$$
- Else:
$$R^2 = RR, \quad S_1 = R + R^2, \quad S_2 = I + R^2, \quad T = S_1 S_2$$
$$X \leftarrow X + 0.5XT$$

Stopping Criterion:

$$\|I - X^\dagger X\|^2 < \epsilon^2$$

2 Halley's Method

Goal: Higher-order iterative refinement with cubic convergence.

Update Rule:

$$U_{n+1} = U_n \left(3 + U_n^\dagger U_n\right) \left(1 + 3U_n^\dagger U_n\right)^{-1}$$

Stopping Criterion: Same as Newton's:

$$\|I - U^\dagger U\|^2 < \epsilon^2$$

3 Cayley-Hamilton Projection

Idea: Use the Cayley-Hamilton theorem to approximate matrix functions with finite polynomials.

Polar Projection:

$$U = \frac{1}{\sqrt{X^\dagger X}} X$$

Avoids direct inverse or square roots via Cayley-Hamilton-based expansion.

Pros: Explicit, non-iterative.

Cons: Less accurate for large matrices.

4 Special Unitary Projection (SU(N))

Constraint: $\det(U) = 1$

Newton Variant

Normalize determinant at each iteration:

$$X \leftarrow X / \det(X)^{1/N}$$

Then proceed with Newton's method.

Exponential Map Projection

Define:

$$A = XX^\dagger - X^\dagger X$$

Compute update:

$$X \leftarrow e^{cA} X$$

where scalar c is chosen to reduce deviation from unitarity.

Summary Table

Method	Iterative	SU(N)	Accuracy	Speed	Notes
Newton	Yes	Yes	High	Fast	Simple, general-purpose
Halley	Yes	Yes	Very High	Medium	Cubic convergence
Cayley-Hamilton	No	No	Medium	Fast	Explicit, no iterations
Exponential Map	Yes	Yes	High	Slow	Only for SU(N)

References

- J.C. Osborn, C.T. Peterson (source code)
- Golub & Van Loan, *Matrix Computations*
- Higham, *Functions of Matrices: Theory and Computation*